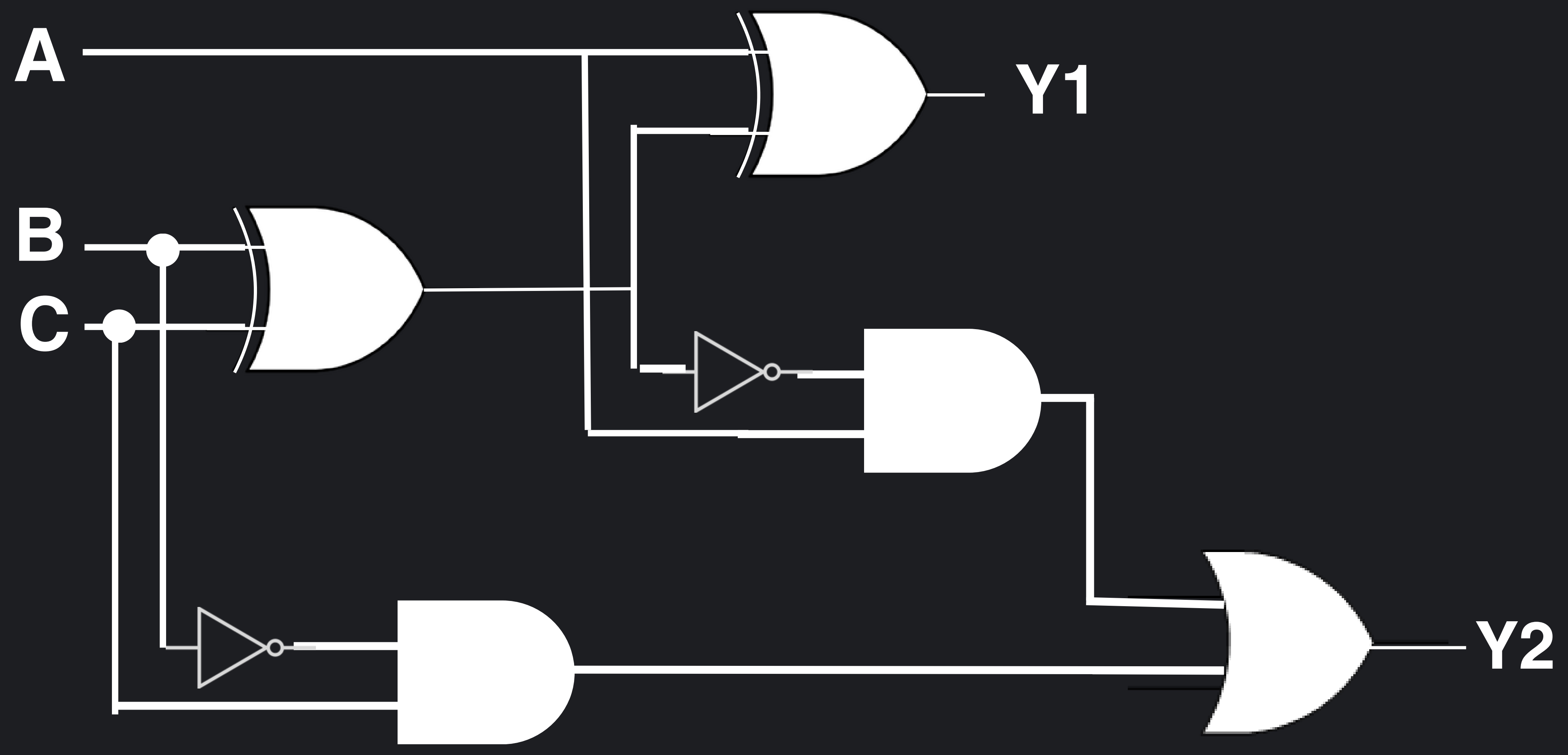
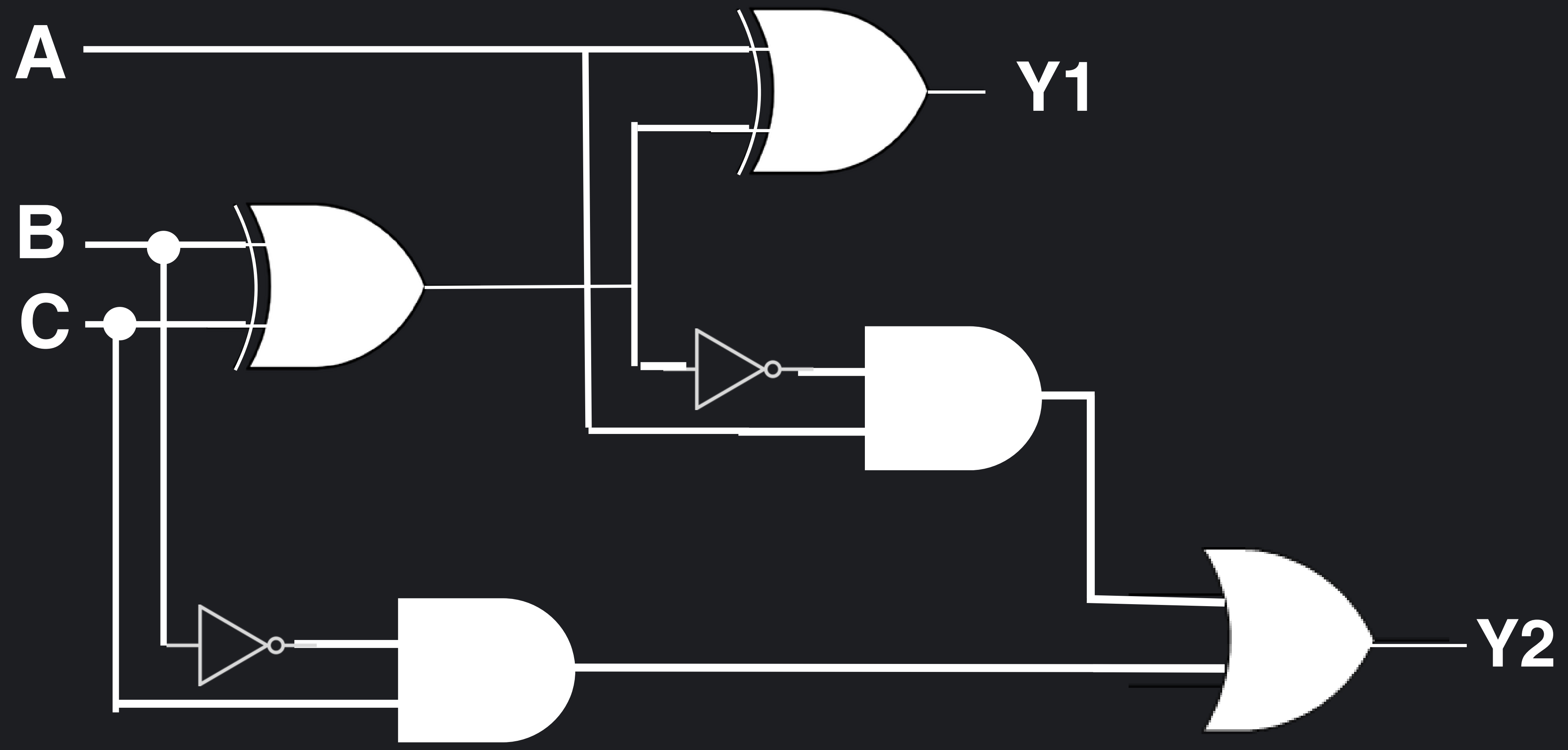
FULL SUBTRACTOR



What is a FULL SUBTRACTOR?

A Full Subtractor is a basic combinational logic element that performs the arithmetic subtraction of three binary bits. It produces two outputs, Difference and Borrow, based on its three inputs which consist of the minuend, the subtrahend, and a Borrow-in from a previous stage.

Truth Table



A	B	C	Y1	Y2
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Function of the FULL SUBTRACTOR

A Full Subtractor performs a logical arithmetic subtraction on its three binary inputs.The main function of a Full Subtractor is to serve as a basic arithmetic element and produce an output signal (Difference and Borrow Out) according to this rule:

Output Difference = 1 when the total of inputs (A, B, Bin) is 1 or 3.

Output Difference = 0 when the total of inputs is 0 or 2.

Output Borrow Out = 1 when the minuend (A) is less than the sum of (B + Bin)

Output Borrow Out = 0 when the minuend (A) is greater than or equal to the sum of (B + Bin).